

Week 7 Discussion Problems Solutions

1. Use Gauss' law to determine the enclosed charge.

$$\Phi_e = \frac{Q_{enc}}{\epsilon_0} \rightarrow Q_{enc} = \Phi_e \epsilon_0 = (1840 N.m^2/C)(8.85 \times 10^{-12} C^2/N.m^2) = 1.63 \times 10^{-8} C \quad (1)$$

2. (a) The electric field points radially out everywhere.
 (b) By symmetry, $\mathbf{E}(\mathbf{r}) = E(r)\hat{\mathbf{r}}$. Take a cylindrical Gaussian surface of height h and radius r . The charge enclosed by it is $Q_{enc} = \rho V = \rho\pi R^2 h$. Then, by Gauss's law,

$$E(r)2\pi rh = \frac{\rho\pi R^2 h}{\epsilon_0}, \implies E_{out}(r) = \frac{\rho R^2}{2\epsilon_0 r}. \quad (2)$$

- (c) For $r < R$, the approach is the same, but now the charge enclosed depends on the radius of the Gaussian surface, $Q_{enc} = \rho\pi r^2 h$. Then, by Gauss's law,

$$E(r)2\pi rh = \frac{\rho\pi r^2 h}{\epsilon_0}, \implies E_{in}(r) = \frac{\rho r}{2\epsilon_0}. \quad (3)$$

- (d) When $r = R$,

$$E_{out} = \frac{\rho R}{2\epsilon_0} = E_{in}, \quad (4)$$

i.e. the electric field is continuous at the surface.

3. Solution:

By symmetry, $\mathbf{E} = E(r)\hat{\mathbf{r}}$. We can immediately say that $\mathbf{E}_A = 0$ because there it's the only symmetric possibility. Since D is inside a conductor, $\mathbf{E}_D = 0$ as well. At point B , we use Gauss's law for a (non-conducting) cylinder and find

$$\mathbf{E}_B = \frac{\rho r_B}{2\epsilon_0} \hat{\mathbf{r}}.$$

At point C , Gauss's law for a non-conducting cylinder (or, equivalently, a wire, since we're outside) gives

$$\mathbf{E}_C = \frac{\rho R_1^2}{2\epsilon_0 r_C} \hat{\mathbf{r}}.$$

The conductor has no electric field inside it, so it must have charge density

$$\sigma_I = -\frac{\rho R_1^2}{2R_2}$$

spread out over its inner surface to cancel out the $\mathbf{E}$ -field. This means it must have an equal and opposite amount of charge on the outer surface, which gives a surface charge density of

$$\sigma_O = \frac{\rho R_1^2}{2R_3}.$$

Outside the conductor, at point E , we don't even see the conductor because the electric field resumes outside; the charge enclosed by a Gaussian surface of radius r_E is the same charge enclosed by a surface with radius r_C . Therefore

$$\mathbf{E}_E = \frac{\rho R_1^2}{2\epsilon_0 r_E} \hat{\mathbf{r}}.$$

4. (a) Increases linearly by Gauss's law: $\mathbf{E} = \frac{3Q}{4\pi\epsilon_0 a^3} r \hat{\mathbf{r}}$
 (b) Like point charge: $\mathbf{E} = \frac{3Q}{4\pi\epsilon_0 r^2} \hat{\mathbf{r}}$
 (c) $\mathbf{E} = 0$ inside a conductor

(d) $\mathbf{E} = \frac{2Q}{4\pi\epsilon_0 r^2} \hat{r}$

(e) Inner surface: $-3Q$, outer surface: $+2Q$

5. To begin, we will find the electric field at some radial distance $r < R$ from the cylinder with no hole is given by Gauss' Law. Take a Gaussian cylinder of length l parallel to the axis. Then we have

$$\oint \mathbf{E} \cdot d\mathbf{A} = 2\pi r l E = \frac{Q_{enc}}{\epsilon_0} = \frac{\pi r^2 l \rho}{\epsilon_0} \rightarrow \mathbf{E} = \frac{r\rho}{2\epsilon_0} \hat{r}. \quad (5)$$

To account for the spherical hole, we recognize that a sphere of radius $R/2$ with charge density $-\rho$ positioned at the same location would create a net charge density of 0 (a spherical hole). To find the electric field created by such a sphere, we consider spherical shells of radius s centered on the negatively charged sphere. Using Gauss' law, when $s \geq R/2$, we have

$$\oint \mathbf{E} \cdot d\mathbf{A} = E(4\pi s^2) = \frac{Q_{enc}}{\epsilon_0} = \frac{4}{3}\pi \left(\frac{R}{2}\right)^3 (-\rho/\epsilon_0) \rightarrow \mathbf{E} = -\frac{\left(\frac{R}{2}\right)^3 \rho}{3\epsilon_0 s^2} \hat{s}. \quad (6)$$

Similarly, when $s < R/2$, we have

$$\oint \mathbf{E} \cdot d\mathbf{A} = E(4\pi s^2) = \frac{Q_{enc}}{\epsilon_0} = \frac{4}{3}\pi s^3 (-\rho/\epsilon_0) \rightarrow \mathbf{E} = -\frac{\rho s}{3\epsilon_0} \hat{s}. \quad (7)$$

Now we can use the principle of superposition to compute the electric field at each point. At point A, $r = 0$ and $s = R/2$. Also, $\hat{s} = -\hat{x}$. Therefore,

$$\mathbf{E}_A = \frac{\left(\frac{R}{2}\right)^3 \rho}{3\epsilon_0} \hat{x} \quad (8)$$

At point B, $r = R/2$ and $s = R/2$. Also, $\hat{r} = \hat{x}$ and $\hat{s} = \hat{y}$. Therefore,

$$\mathbf{E}_B = -\frac{\left(\frac{R}{2}\right)^3 \rho}{3\epsilon_0} \hat{y} + \frac{R\rho}{4\epsilon_0} \hat{x} \quad (9)$$

At point C, $r = R/2$ and $s = 0$. Also, $\hat{r} = \hat{x}$. Therefore,

$$\mathbf{E}_C = \frac{R\rho}{4\epsilon_0} \hat{x} \quad (10)$$

6. (a) Consider a Gaussian cylinder of length l and radius r . On the left hand side of Gauss's law, we have

$$\oint \mathbf{E} \cdot d\mathbf{A} = E(2\pi r l). \quad (11)$$

On the right hand side, we have

$$\frac{q_{enc}}{\epsilon_0} = \int_0^l \int_0^{2\pi} \int_0^r \rho_0 \left(a - \frac{r}{b}\right) r dr d\theta dz \quad (12)$$

$$= \frac{\rho_0}{\epsilon_0} \left(\frac{ar^2}{2} - \frac{r^3}{3b}\right) 2\pi l \quad (13)$$

So,

$$E = \frac{\rho_0 r}{\epsilon_0} \left(\frac{a}{2} - \frac{r}{3b}\right) \quad (14)$$

(b) The left side of the equation remains the same. The right hand side becomes

$$\frac{q_{enc}}{\epsilon_0} = \int_0^l \int_0^{2\pi} \int_0^R \rho_0 \left(a - \frac{r}{b}\right) r dr d\theta dz \quad (15)$$

$$= \frac{\rho_0}{\epsilon_0} \left(\frac{aR^2}{2} - \frac{R^3}{3b} \right) 2\pi l \quad (16)$$

So,

$$E = \frac{\rho_0 R^2}{\epsilon_0 r} \left(\frac{a}{2} - \frac{R}{3b} \right) \quad (17)$$

7. Solution:

We will use superposition. Consider one small arc of the circle subtended by the angle $d\theta$. The infinitesimal charge on that arc is $dq(\theta) = \lambda(\theta)d\ell = \lambda_0 \cos^2 \theta R d\theta$. Then

$$V(z) = \int_0^{2\pi} \frac{k dq}{r} = \int_0^{2\pi} \frac{k \lambda_0 R \cos^2 \theta d\theta}{\sqrt{R^2 + z^2}} = \frac{k \lambda_0 R}{\sqrt{R^2 + z^2}} \int_0^{2\pi} \cos^2 \theta d\theta = \frac{k \lambda_0 R}{\sqrt{R^2 + z^2}} \pi.$$